HW #2 — Axioms of Probability; Cond. Probability (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Friday, 9/18 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as two files: one with all textbook and non-R problems as a .pdf document, and one that is rendered .pdf file from the attached Quarto document. Any submissions not in this format are subject to point deductions.
- Responses can be hand-written or typed for the non-R problems.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

Background

The second homework will look at topics of counting, the axioms of probability, and conditional probability. You will also be using a Monte Carlo simulation to replicate a previous HW problem.

Part I: Textbook Problems (15 pts)

From *Introduction to Probability, Second Edition*:

- Chapter 1: Q38, Q44
- Chapter 2: Q3, Q11, Q35

Part II: Quality Control (5 pts)

A manufacturer of snow shovels claims that their shovels will last 80% of the time after 10 hours of use. An independent tester tries to verify this claim by using many of the same shovel for a single hour on different surfaces. They note that:

- The probability of that manufacturer's shovel being defective initially is 10%.
- The probability of that manufacturer's shovel being fine after one hour of usage, assuming it is not defective initially is 98%.
- The probability of shoveling snow on a soft surface is 20%, a hard surface 65%, and any other surface 15%.
- The probability of shoveling snow on a soft surface given that the manufacturer's shovel is fine after one hour is 15%.
- Assume wear happens uniformly over time, meaning that the shoveling that happens in one hour will not have any effect on its performance in the next hour.
    - See memoryless property for more details.

Problems to Answer:

(a) Calculate the probability of having a working shovel that is not defective after one hour of use.

(b) Calculate the probability of having a working shovel after one hour of use.

- *Note*: If I have a defective shovel, will it even work in the first place?

(c) Calculate the probability of having a working shovel after one hour given that we are using it on a soft surface.

(d) Calculate the probability of having a working shovel that is not defective after 10 hours of use.

- *Note*: Recall the last bullet point.

(e) Calculate the probability of having a working shovel after 10 hours of use, given it is not defective.

- *Note*: Recall the last bullet point.

(f) Determine if the claim holds up, using information given and calculated up until this point.

Part III: Monte Carlo Simulation (5 pts)

At times, you may find that the probabilities you are trying to solve for are rather complicated. This can be a result of complex setups, incomputable calculations by-hand, or maybe a lack of will power to solve.

Regardless of the reason, we can use Monte Carlo simulations to solve for these questions. We harness the power of random sampling and many, many experiments to get empirical estimates of the probabilities that we want to find. To run the simulations, we must do the following:

1. Set up a model, in which we identify possible outcomes or independent variables, as well as the ultimate probability. We want to also determine how we would map from the outcomes to that probability.
2. Determine a probability distribution of these outcomes.
3. Perform a random sample, and repeat this process repeatedly for some N times.

You can think of a Monte Carlo simulation as us trying to repeat the same situation hundreds of thousands of times and trying to count how many times the event we care about occurs. A Monte Carlo simulation works here because a computer can perform these trials over and over again really quickly.

For example, suppose I care about the chance of getting two pairs in a game of poker. You can see an example of how a Monte Carlo simulation is set up in the R code bulleted under the HW assignment on my website. We do the following:

1. Set up a model: We list all possible outcomes. I use a vector of 1 through 13, repeated 4 times over. For this situation, we do not care about suit, so we do not need to account for this. Additionally, note that if we select five cards, we hope to find 2 groups of numbers that have counts of 2.
2. Determine a probability distribution of these outcomes: The `sample` function comes in handy here. We can use it to draw from our outcomes 5 times over. We have an equal chance of drawing one of these 52 cards. So in our sample function, we do not set a `probs` argument here. However, if there are unequal probabilities of those outcomes, we would specify the probabilities as a vector in order.
3. Perform a random sample, and repeat this process repeatedly for some N times: This is where the `sample` function and for-loop come into play. We run the `sample` function and identify if we have two pairs. If so, then great: we add 1 to our counter. Otherwise, we repeat the experiment N times using a for-loop.

After the N draws, we can find the empirical probability, where we take the number of times the hand happened over the number of draws, and tah-dah, we have our result! However, note that this is an estimate, and our estimate may be off within some standard error away from the true result.

Now, it's your turn. We will re-do Part III from the last homework using a Monte Carlo simulation. Use the code on the website as a template, and add code to it with answers here to complete the following questions:

Problems to Answer:

(a) In the code, fill in the blanks seen in lines 38, 46, and 49. Remove the blanks and pound signs after doing so.

(b) Run all code from lines 31-55 by highlighting code and hitting run. What value do you get?

(c) Repeat part (b) again 9 more times. What values do you get?

(d) Change N to 1000 and 10000. What happens to the value seen in line 55?

(e) Now run line 58. What do you notice happens to the empirical probability as we increase N?